ANSWER SHEET DAC 2025

PRELIMINARY ROUND



Team Name : sayonarasrama [-1]

Himpunan Mahasiswa Statistika ITS (HIMASTA-ITS)
Gedung H Lantai III, Jl. Arief Rahman Hakim
Kampus ITS Sukolilo Surabaya
Email: pekanrayastatistika2025@gmail.com
Contact Person:
Hita (+62 857-1207-7572)
Ali (+62 812-3590-7687)

CHAPTER I: Introduction

A. Background

Large Language Model (LLM) apps such as GPT, Claude, Gemini, Perplexity, Grok, and Deepseek have rapidly expanded into mainstream use, powering applications from conversational assistants to productivity software and educational platforms. In 2025, GPT leads with 400 million active weekly users, dominating the LLM app market with billions of monthly visits; Gemini follows with around half that user base, while Claude, Grok, Perplexity, and Deepseek maintain smaller but rapidly growing shares, collectively reflecting widespread adoption across diverse sectors worldwide [1–2]. Alongside this growth, user reviews have become a vital source of real-world feedback. Unlike usage logs or quantitative performance benchmarks, reviews capture nuanced sentiment signals such as trust, satisfaction, frustration, and skepticism that reflect authentic user experiences [3–5]. As these applications evolve, developers increasingly rely on such feedback to identify strengths, benchmark against competitors, and guide product improvements.

A core challenge lies in systematically classifying sentiment expressed in reviews. While positive reviews dominate due to reporting bias [6], negative reviews often provide the most actionable insights, highlighting issues such as slow response times, factual inaccuracies, or confusing interfaces [7]. Yet, these reviews are typically short, informal, and context-dependent, making automated analysis difficult. Effective sentiment classification systems must therefore handle noisy, imbalanced data while ensuring fair and reliable categorization across positive, neutral, and negative classes.

Beyond classification, analyzing sentiment across application versions provides critical longitudinal insights. Although version updates are designed to introduce improvements, they may also lead to regressions or unintended problems that degrade user experience. Tracking sentiment over time enables developers to detect shifts in perception, compare user reception across releases, and prioritize improvements where dissatisfaction clusters around specific versions [8–9]. Against this backdrop, the present study aims to integrate sentiment classification with cross-version analysis to generate actionable insight for LLM developers.

To address these challenges, this study proposes the development of a transformer-based sentiment classification model tailored for user reviews of LLM applications. Beyond sentiment polarity, we employ topic modeling technique to extract and cluster the underlying issues users frequently mention, enabling more granular identification of pain points. Statistical tests such as chi-square are further applied to examine shifts in issue prevalence and sentiment proportions across application versions. Together, this integrated framework provides developers with longitudinal insights to track regressions, prioritize improvements, and evaluate the evolving reception of LLM apps.

B. Research Questions

Based on the background above, this study is guided by the following research questions:

- RQ1: How accurately and fairly can sentiment classification models capture nuanced and imbalanced user reviews of LLM applications?
- RQ2: What key themes and recurrent issues can be uncovered from user reviews, particularly negative ones, using topic modeling approaches?
- RQ3: How do user sentiments and issue distributions shift across application versions, and to what extent can statistical analysis systematically detect these changes?

These questions guide the study toward building a system that not only classifies sentiment accurately but also extracts insights from negative reviews and tracks shifts across app versions.

CHAPTER II: Theoretical Framework

A. Text Representation

Recurrent Neural Networks (RNN) [10] and Long Short-Term Memory (LSTM) [11] have been widely used for natural language processing (NLP) tasks. However, RNNs suffer from training difficulties on long sequences due to challenges in maintaining contextual information over extended text. LSTMs alleviate this with their gating mechanisms that enable long- and short-term memory, but their computational complexity can limit scalability and sometimes results in lower performance..

In recent years, transformer-based models have revolutionized NLP by employing self-attention mechanisms that capture complex relationships between words in parallel, enabling efficient modeling over large-scale text [12]. More recently, cutting-edge models like E5 Multilingual [13] and Qwen3 Embedding [14] have pushed the state-of-the-art in text representation and language understanding even further. These models leverage advanced training techniques and architectures to consistently achieve top performance across diverse benchmarks, most notably the Massive Text Embedding Benchmark (MTEB) [15], which evaluates embeddings on a wide range of retrieval, clustering, and classification tasks. Their strong and robust results on MTEB and other evaluations set new standards for both general-purpose and multilingual language models.

E5 (**E**mb**E**ddings from bidir**E**ctional **E**ncoder r**E**presentations) model generates embeddings by encoding input text into dense vector representations using a transformer encoder architecture optimized for semantic similarity tasks. Instead of treating text as isolated tokens, E5 is trained with weakly supervised contrastive learning objectives, where semantically related inputs (e.g., query–passage pairs) are pulled closer in vector space, while unrelated pairs are pushed apart, as shown in Figure 2.1.

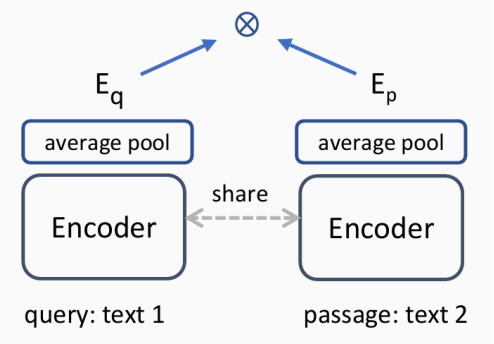


Figure 2.1. The E5 architecture, where Ep (passage embedding) and Eq (query embedding) are compared for similarity in contrastive learning [13].

This training paradigm, combined with large-scale multilingual corpora, allows E5 model to capture fine-grained semantic relationships across languages and domains. As a result, its embeddings are highly effective for downstream applications such as clustering, semantic search, topic modeling, and sentiment analysis, making it a robust backbone for real-world NLP tasks.

B. Gradient Boosting Tree Model

Gradient Boosting Machine (GBT) is an ensemble learning technique that builds models sequentially to correct errors made by previous models. It uses boosting, where each new model aims to improve upon the previous ensemble by correcting its errors. GBT employs a gradient-descent approach in function space to optimize the model [16].

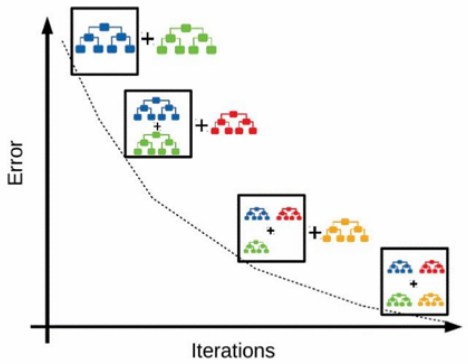


Figure 2.2. Iteration–error–new model relationship [17]

As machine learning advanced, more efficient variants of GBT such as XGBoost [18] and LightGBM [19] emerged, offering faster training and improved accuracy through optimized regularization techniques, subsampling, and reduced computational complexity. However, these variants still generally require explicit transformations to handle categorical features, such as one-hot encoding or label encoding, which can increase data dimensionality and the risk of overfitting. CatBoost (Categorical Boosting) [20] was developed to address these limitations by enabling the direct processing of categorical features through techniques such

as ordered boosting and target statistics. The algorithm also applies symmetric tree building to accelerate prediction and incorporates built-in regularization to reduce the risk of overfitting.

C. Transfer Learning for Text Classification Modeling

In text classification, models usually consist of two main parts: a feature extractor and a classifier. The feature extractor transforms raw text into numerical representations (embeddings), while the classifier maps these embeddings into the target labels. The better the text representation, the more accurate the classification results can be [21]. For a new task, building good text representations from scratch is often difficult because of limited data and resources. This is where transfer learning becomes useful. Transfer learning means reusing knowledge from a large pre-trained model and adapting it to a specific task. This allows models to take advantage of rich text embeddings without needing to train everything from zero, making them more effective and efficient [21].

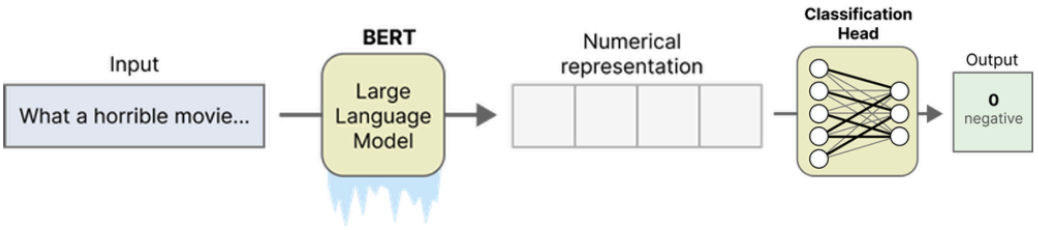


Figure 2.3. Classic architecture for text classifier model [21]

In practice, transfer learning is usually applied by freezing the encoder (the feature extractor) and only training the classifier head on top of it as illustrated in Figure 2.3. The classifier can be a simple artificial neural network (ANN) with fully connected layers or a gradient boosting model such as XGBoost, LightGBM, or CatBoost. This way, the model benefits from high-quality pre-trained embeddings while keeping the training process efficient.

D. Topic Modeling: BERTopic

Topic modeling is an unsupervised learning technique used to automatically discover hidden themes or topics within a collection of documents. Instead of manually labeling text data, topic modeling algorithms group words or sentences that frequently occur together, making it possible to summarize large text corpora into interpretable clusters of topics. The main purpose of topic modeling is to uncover the underlying structure of text data. Its use cases are widespread, including analyzing customer feedback, extracting themes from social media discussions, organizing research articles, and supporting recommendation systems. By identifying topics, organizations and researchers can gain insights into patterns, trends, and sentiments within unstructured text datasets.

BERTopic builds on this idea by generating topic representations through three main steps as illustrated in Figure 2.4.

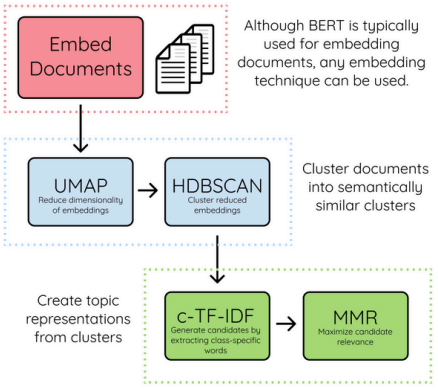


Figure 2.4. BERTopic concept illustration

First, each document is transformed into an embedding using a pre-trained transformer-based language model, which captures rich semantic information. Second, the dimensionality of these embeddings is reduced to make the clustering process more efficient and effective. Third, the clustered documents are used to derive interpretable topics through a class-based variation of TF-IDF, which highlights representative words for each cluster. By combining deep contextual embeddings, dimensionality reduction, and class-based TF-IDF, BERTopic is able to produce coherent and competitive topic representations compared to both classical and modern clustering-based approaches [22].

E. Chi-Square Test of Independence

The Chi-Square Test of Independence is a non-parametric statistical test used to determine whether there is a significant association between two categorical variables. Unlike tests designed for numerical data, the Chi-Square test works with frequency counts in contingency tables, making it widely applicable in analyzing survey data, experimental results, and categorical observations.

The procedure begins with constructing a contingency table, which displays the frequency distribution of variables in rows and columns. The test compares the observed frequencies in each cell of the table with the expected frequencies that would occur if the two variables were completely independent. The Chi-Square statistic is then computed as:

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where O_{ij} is the observed frequency and E_{ij} is the expected frequency in cell i, j.

In summary, the Chi-Square Test of Independence is a fundamental statistical tool

for examining relationships between categorical variables, providing a simple yet powerful method to test for independence in categorical data analysis.

F. Class Weighting in Loss Function

In imbalanced classification tasks, models tend to be biased toward majority classes, resulting in poor performance for minority classes. To address this, class weighting is incorporated into the loss function. For multi-class classification with Cross-Entropy Loss, the weighted formulation is:

$$\mathcal{L}(x, y) = -w_y \cdot \log\left(\frac{e^{z_y}}{\sum_{j=1}^{K} e^{z_j}}\right)$$

where w_y is the weight assigned to the true class y. This adjustment ensures that errors from minority classes contribute more significantly to the overall loss, thus encouraging the model to learn a more balanced representation across all classes.

G. Evaluation Metric: Macro F1 Score

The F1 Score is an evaluation metric that combines Precision and Recall into a single value, providing a balance between the two. It is defined as the harmonic mean of Precision and Recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

where Precision measures how many of the predicted positive samples are actually positive and Recall measures how many of the actual positive samples are correctly identified by the model.

When dealing with multi-class classification, the F1 score can be averaged across classes using the Macro F1 Score. The Macro F1 Score is calculated by computing the F1 score for each class independently and then taking the arithmetic mean of these scores:

$$F1_{macro} = \frac{1}{N}\sum_{i=1}^{N} F1_i$$

where N is the number of classes, and $F1_i$ is the F1 score of class i. This averaging method treats all classes equally, regardless of their frequency in the dataset. As a result, F1-Macro is especially useful when the dataset is imbalanced, since it ensures that minority classes contribute equally to the final score.

CHAPTER III: Analytical Steps

The following Figure 3.1 shows the diagram of the data analysis steps to be carried out.The following subsections will provide a more in-depth explanation.

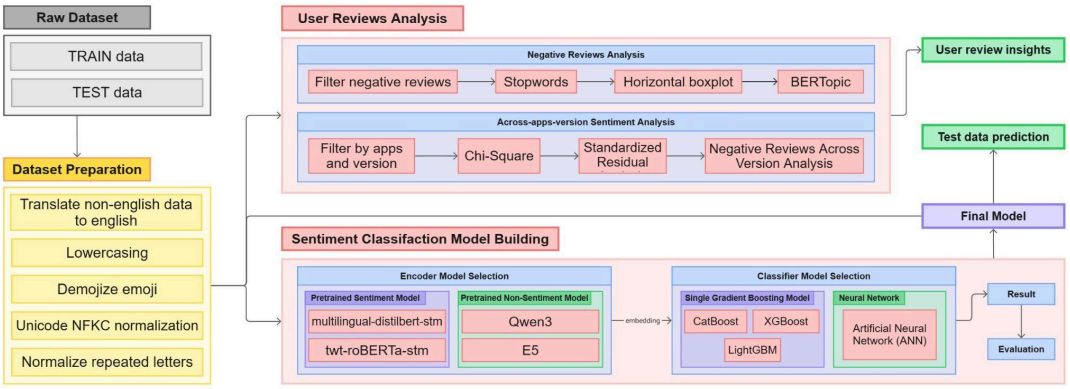


Figure 3.1. Analytical steps diagram

A. Raw Dataset

The dataset used in this study was provided by the PRS ITS 2025 committee through the following link. It was divided into two parts: the training dataset and the testing dataset. The training dataset was utilized to build and refine the sentiment classification model as well as to support the analysis of user reviews. In contrast, the testing dataset was reserved for the final evaluation stage, aiming to objectively assess the model's performance and ensure that it generalizes effectively to unseen data. The dataset itself contains several key attributes, including Comment (user review text), At (timestamp of the review), AppVersion (application version at the time of the review), and Sentiment (only in training dataset). A snapshot of the dataset is provided in Attachment 1 to illustrate the structure and content of the data.

B. Dataset Preparation

Before any analysis can be performed, the raw dataset undergoes a series of preprocessing steps to clean and normalize the text. The following Table 3.1 presents the data preprocessing steps applied to the *'Comment'* column.

Table 3.1. Illustration of data transformation after preprocessing

Preprocessing Step	Initial Comment	Transformed Comment
Translate non-english comment to english	أغبى نموذج لغوي للمستخدمين غير الناطقين باللغة الإنجليزية.	*The dumbest language model for non-English speaking users.*
Lowercasing	*THE BEST!!!!!!!!!*	*the best!!!!!!!!!*
Demojize emoji	❤️🤍	*:red_heart::white_heart:*

Unicode NFKC normalization	*freat app you can use it for anething*	*great app you can use it for anything*
Normalize repeated letters	*it isssss sooooo goooood*	*it is so good*

As we can see, non-English reviews are translated into English using GoogleTranslator from the deep-translator package to maintain language consistency. All text is converted into lowercase to avoid case-related mismatches. Emojis are demojized into their textual equivalents to preserve their meaning in textual form. Unicode NFKC normalization is applied to standardize characters that may have visually similar but technically different representations, such as full-width digits or special symbols. Repeated letters are normalized to ensure consistency across variations in spelling. Finally, duplicate entries are removed to avoid bias in the model. These preparation steps help reduce noise in the data.

C. Sentiment Classification Model Building

The process of building the sentiment classification model in this study follows a structured pipeline. First, each review text is transformed into a numerical representation by generating vector embeddings using a pre-trained encoder model. Several encoder models from HuggingFace's open-source repository (see Table 3.2) are compared to identify which one produces the most effective text representation.

Table 3.2. Encoder model checkpoint

Model Name	Model Checkpoint
multilingual-distilbert-stm	tabularisai/multilingual-sentiment-analysis
twt-roBERTa-stm	cardiffnlp/twitter-roberta-base-sentiment-latest
Qwen3/0.6B	Qwen/Qwen3-Embedding-0.6B
Multilingual E5	intfloat/multilingual-e5-large

Once embeddings are generated, they are fed into a classifier. Two types of classifiers are employed: boosting-based models (CatBoost, XGBoost, LightGBM) and a feed-forward Artificial Neural Network (ANN). The ANN architecture used in this study is described in Table 3.3.

Table 3.3. ANN architecture in this study

Layer	Detail
Input Layer	Dimension = *input_dim* (embedding size)

Hidden Layer 1	Fully Connected (Linear, 512 units) + ReLU + Dropout (0.3)
Hidden Layer 2	Fully Connected (Linear, 256 units) + ReLU + Dropout (0.3)
Output layer	Fully Connected (Linear, *num_classes*)

For model evaluation, the dataset is split into training and validation sets, with 20% of the training data reserved for validation using stratified sampling to maintain class distribution. The evaluation metric employed is the Macro F1-score. This ensures that the performance is measured fairly across both majority and minority classes.

D. User Review Analysis

At this stage, the analytical process branches into two complementary sub-analyses, each designed to capture different but equally important dimensions of user reviews.

a. Negative Review Analysis

In conducting negative review analysis, the first step is to remove stopwords. Stopwords do not carry meaningful information about customer complaints, so they are excluded to highlight more informative words such as nouns, verbs, or adjectives.

Once the text is cleaned from these common words, a frequency analysis is performed to identify the most recurring keywords in the negative reviews. These keywords are then visualized using a horizontal barplot, making it easier to see which themes of complaints are most dominant. However, frequency alone does not provide the full picture. Therefore, the next step is to examine the specific review lines that contain those keywords to gain deeper contextual understanding. Topic modeling is also conducted to get broader insights. This helps clarify whether *delay* refers to shipping issues, whether *service* points to poor customer support, or whether *expensive* reflects customer dissatisfaction with pricing.

By combining the quantitative insight from the barplot and with the qualitative exploration of context and topic modeling exploration, a more comprehensive view of customer dissatisfaction can be obtained. This approach not only reveals what customers often complain about but also explains why those complaints arise, providing a stronger foundation for targeted improvements.

b. Across-apps-version Sentiment Analysis

For this part, reviews are grouped by their respective *AppVersion*, enabling a comparative analysis across different releases of the application. Sentiment distributions are computed for each version to capture shifts in user perception

over time. Visualizations such as stacked bar charts are generated to reveal trends and anomalies across versions.

To assess whether the observed differences are statistically significant, a Chi-Square test of independence is performed between sentiment and app versions. Following this, standardized residuals are examined to identify which specific app versions exhibit unusual sentiment proportions compared to expectations.

Finally, for versions that stand out, a negative review keyword analysis is conducted. Using horizontal barplots, the most frequent complaint-related terms are extracted and contextualized to determine whether shifts in user dissatisfaction point to new or recurring issues tied to specific releases of each LLM application.

CHAPTER IV: Analysis of Results

A. Sentiment Classification Modeling Results

The analysis begins with an inspection of the dataset, which reveals a pronounced class imbalance across sentiment categories as shown Figure 4.1 below.

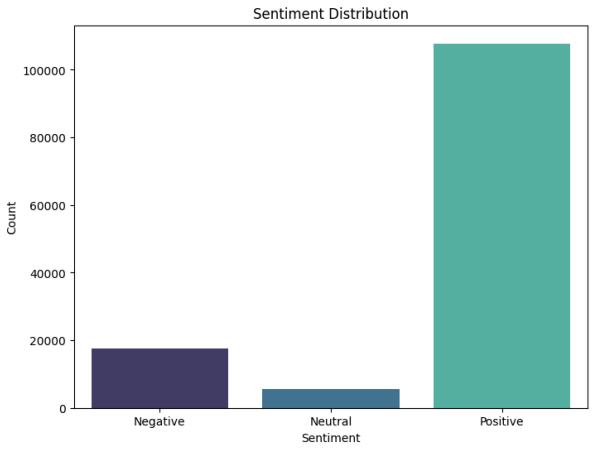


Figure 4.1. Sentiment class distribution in training dataset

Specifically, the dataset contains 107771 positive reviews, 17520 negative reviews, and only 5589 neutral reviews. Such a distribution highlights the dominance of the positive class and the severe underrepresentation of the neutral class. This imbalance motivates the use of class weighting strategies [Section II.F] during model training to prevent the classifier from being biased toward the majority class and to ensure that minority classes are adequately represented in the learning process.

In the initial phase, we conducted a series of experiments to jointly compare different encoder–classifier combinations in order to identify the most effective architecture for sentiment classification. For this purpose, the training dataset was partitioned into training and validation subsets with stratify method, ensuring

proportional representation of sentiment classes. The validation subset was used to monitor model performance and guide selection. The comparative results are summarized in Table 4.1.

Table 4.1. Model building experiment results: Macro F1 Scores

Encoder Model	LightGBM	XGBoost	CatBoost	ANN
multilingual-distilbert-stm	0.64	0.64	0.62	0.64
twt-roBERTa-stm	0.66	0.66	0.65	0.66
Qwen3/0.6B	0.64	0.63	0.63	0.63
Multilingual E5	**0.66**	**0.67**	**0.66**	**0.66**

From the table above, the best performance is achieved by Multilingual E5 with XGBoost (Macro F1 = 0.67). Overall, Multilingual E5 consistently outperforms other encoders, highlighting the importance of embedding quality. Interestingly, lightweight sentiment-specialized models such as twt-roBERTa-stm also perform competitively, even rivaling the more resource-intensive Qwen3/0.6B, thanks to their prior sentiment knowledge. The results further show that performance differences stem mainly from the encoder choice, while the impact of the classifier is minimal.

To assess the generalization capability of the model, two complementary approaches were undertaken: benchmarking against external participants and error analysis on the validation set. On the held-out 30% test set, our model achieves a Macro-F1 score of 0.64482, placing it at rank 17 out of 69 submissions. Notably, the gap with higher-ranked systems is minimal (only about 0.01 to 0.03 points) indicating that the model performs on par with several of the top entries. Complementing this, error analysis reveals further insights into prediction behavior. A focused inspection of validation samples labeled as neutral shows that many instances flagged as misclassifications are, in fact, more accurately captured by the model than by the provided ground truth labels when judged manually, as shown in Table 4.2. Check this link for more misclassified samples.

Table 4.2. Misclassified samples correctly predicted upon manual inspection

Comment	Ground Truth Label	Predicted Label	Manual Annotation
this app's messed up; it's running in the background for no reason.	Neutral	Negative	Negative
absolutely good, i really love this new friend gemini :red_heart:	Neutral	Positive	Posirive

best platform	Neutral	Positive	Positive

This highlights the robustness of the system, demonstrating both quantitative competitiveness and qualitative reliability in handling noisy or inconsistent annotations. Together, these evaluations confirm the model's strong generalization performance and practical applicability.

B. Insights from User Reviews Analysis

a. Negative Review Analysis

In this section, we analyze negative reviews to assess how sentiment varies across different app versions. The analysis begins by isolating reviews labeled as Negative (0). Once separated, the dataset is further divided according to the type of LLM to allow for a more detailed inspection of language use across models. After this segmentation, a basic text preprocessing step is applied, where stopwords are removed. This is important because stopwords occur frequently but contribute little semantic value, and retaining them could obscure more meaningful terms. As an initial exploration, horizontal barplot are then generated for each LLM to visualize the most frequent words in negative reviews, offering a quick overview of recurring language patterns. Interestingly, even after stopword removal, several terms remain dominant, including *"app", "bad", "worst", "worse",* the model name itself, as well as polarity-related terms such as *"good," "nice,"* and *"ai."* [Attachment 2]. Although these words are fairly generic, their persistence reflects the direct and emotionally charged style of user feedback in this context. To uncover more informative signals, these domain-generic terms are subsequently filtered out, and the horizontal barplot are regenerated, as shown in Figure 4.2.

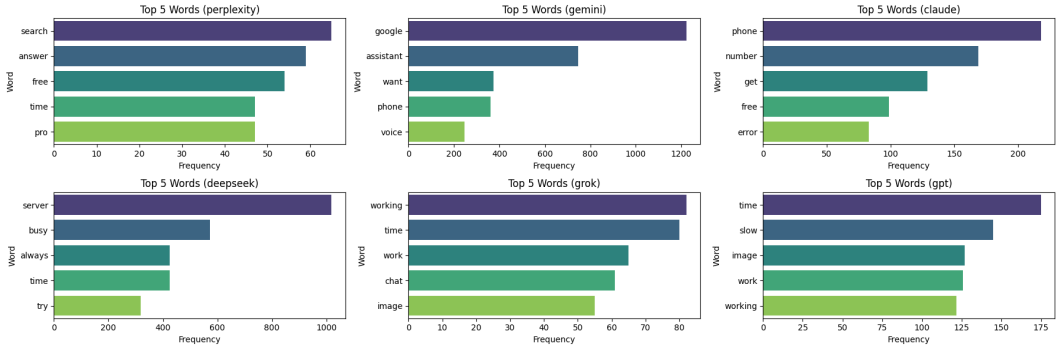
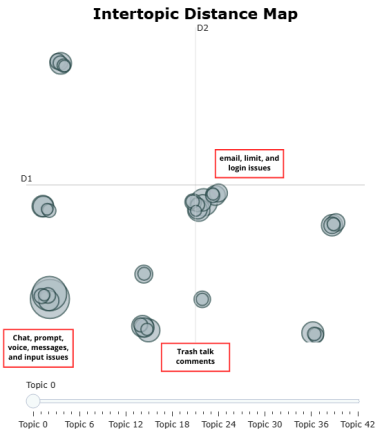


Figure 4.2. Horizontal barplot of most frequent words in negative reviews

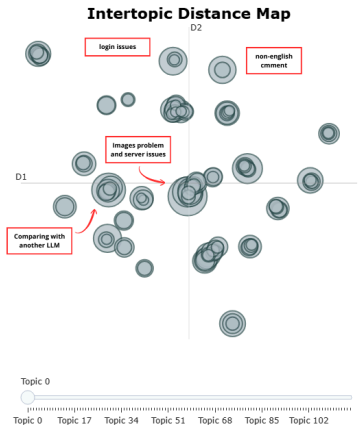
The following insights are derived from both the horizontal barplots above and manual inspection of the contexts in which those words appear [see Attachment 3 for examples]. User complaints reveal distinct issue patterns across LLM apps. Perplexity users express frustration with search-like responses, incomplete answers, and restrictions on free usage perceived as time-wasting. Gemini reviews

focus on dissatisfaction with its role as a Google Assistant replacement, citing reduced functionality, weak device integration, and broken voice commands. For Claude, resistance to mandatory phone number verification dominates, alongside recurring login and registration errors tied to privacy concerns. DeepSeek is heavily criticized for technical instability, with persistent server overloads, failed verifications, and frequent crashes. Grok users highlight persistent chat bugs and disappointment with its image-generation feature, often described as deformed or unusable. Meanwhile, GPT complaints center on slow responses, failed image processing, and recurring update issues across devices. Overall, while the specific complaints differ per application, common issues emerge around reliability, speed, and feature performance, which strongly influence user trust and satisfaction.

While frequency-based inspection offers an accessible overview, it is limited in capturing broader patterns and co-occurring themes. To address this, we apply topic modeling using BERTopic, where embeddings are generated with the *paraphrase-MiniLM-L3-v2* model. The **interactive visualizations** of the resulting topics are saved in HTML format, which can be accessed and downloaded via the following **link**. We annotated topics with most frequency occurrences, with the results shown in Figure 4.3.



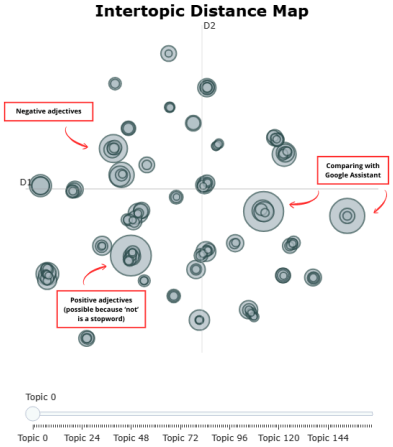
(a.) Claude



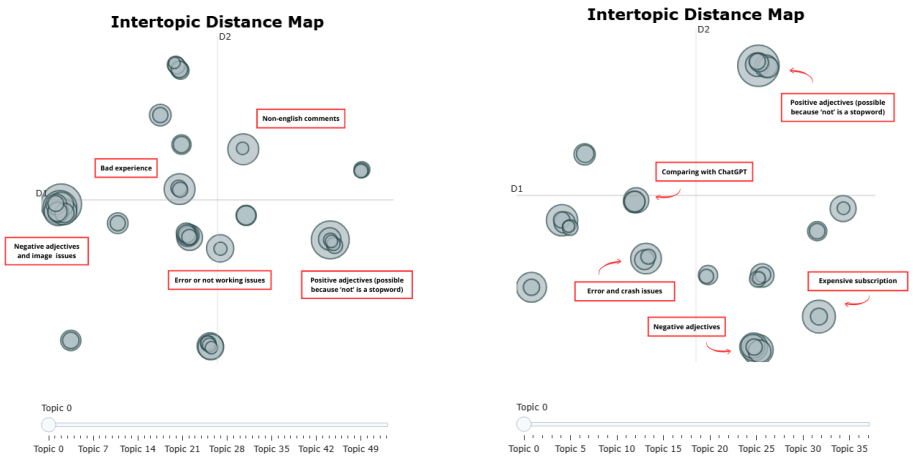
(b.) Deepseek



(c.) GPT



(d.) Gemini



(e.) Grok (f.) Perplexity

Figure 4.3. Static visualization of topic modeling results and our annotation

As can be seen, the topic modeling visualization provides richer and more structured insights compared to simple frequency counts. For example, in the case of Perplexity, several topics clearly revolve around complaints about high subscription costs and the app's tendency to crash or show frequent errors. Another recurring theme is direct comparisons with other LLM applications, where users express disappointment that Perplexity performs worse or offers less value for the price. By analyzing the results of topic modeling, we can obtain a more structured view of user complaints, making it easier to identify priority areas for improvement in each LLM application.

b. Across-apps-version Sentiment Analysis

In this section, we conduct a version-based sentiment analysis by grouping reviews according to *AppVersion* and calculating the proportion of each sentiment class per version. Before proceeding with the main analysis, it is important to note that several missing values were present in the *AppVersion* column.

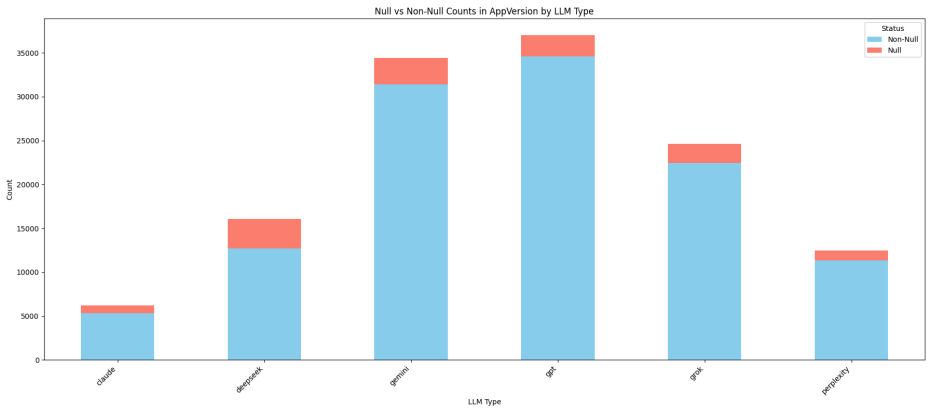


Figure 4.4. Missing values in *AppVersion* column

To address this, we imputed the missing entries using the values from the preceding rows. This decision was based on the pattern shown in Attachment 4, where once reviews were ordered chronologically, neighboring rows tended to share similar app versions.

Before moving to sentiment proportion analysis, we examined the number of reviews available for each app version, as shown in following figure.

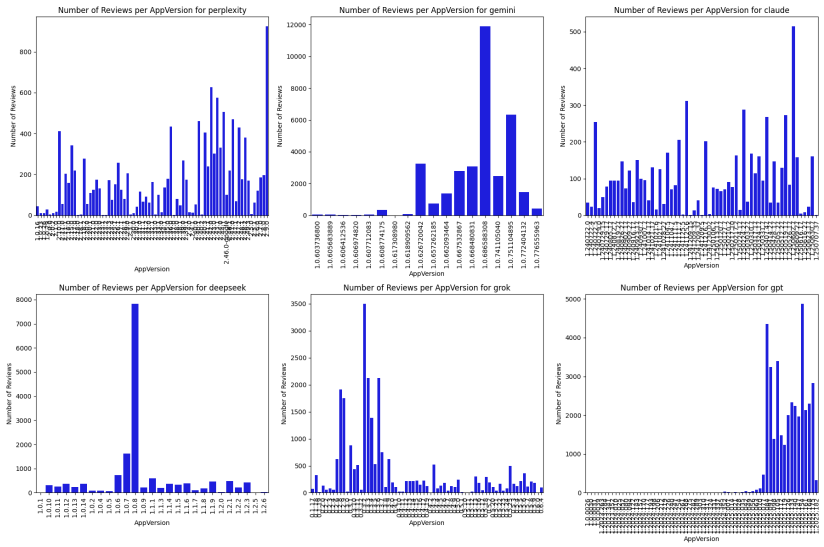


Figure 4.5. Review counts across app-versions

It can be observed that some versions contained only a very small number of reviews. These versions were excluded from the analysis to ensure reliability. Subsequently, the visualization of sentiment proportions across versions for each LLM is presented in Attachment 5. From these visualizations, the most distinct pattern is observed in the Gemini app, where negative sentiment proportion exhibits a declining trend across successive updates, while positive sentiment proportiondisplays a corresponding upward trend.

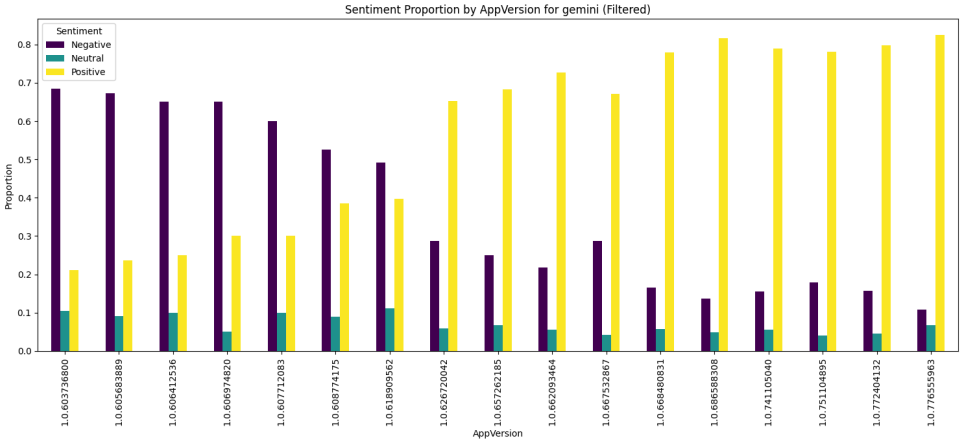


Figure 4.6. Sentiment proportion across app-versions for Gemini

In contrast, the other LLM apps show relatively stable sentiment proportions across versions, with no notable shifts. To substantiate these observations, further statistical tests were conducted.

To examine differences in sentiment proportions across app versions for each LLM, the most appropriate statistical test is the Chi-Square test of independence, as it evaluates whether two categorical variables (in this case, *AppVersion* and *Sentiment*) are associated. The null hypothesis assumes that sentiment distribution is independent of app version, while the alternative hypothesis posits a significant association between the two.

Table 4.3. Chi-Square test of independence results

LLM Apps	Chi-Square Statistics	p-value	Result
Perplexity	225.93	0.0000	H_0 rejected: There is a significant association between *AppVersion* and *Sentiment*
Gemini	1358.91	0.0000	
Claude	268.37	0.0000	
Deepseek	156.42	0.0000	
Grok	956.83	0.0000	
GPT	51.96	0.3225	H_0 accepted: There is no significant association between *AppVersion* and *Sentiment*

As shown in Table 4.3, the Chi-Square test results indicate that for most LLMs (Perplexity, Gemini, Claude, DeepSeek, and Grok), there is a statistically significant relationship between app version and sentiment. This finding implies that the proportions of negative, neutral, and positive sentiment vary meaningfully across different app versions for these models. In contrast, for GPT, the relationship was not statistically significant, suggesting that sentiment proportions remained relatively stable across its versions.

While the Chi-Square test in Table 4.3 confirms that sentiment proportions vary across app versions, it does not reveal *which specific versions* drive these differences. To address this, we conducted a standardized residuals analysis, which compares observed and expected frequencies for each sentiment–version pair. Large positive or negative residuals indicate versions where certain sentiments are disproportionately represented. Identifying these versions is particularly useful, as it helps highlight which app updates may have triggered unusually positive or negative user responses. This provides a more actionable perspective beyond the overall Chi-Square results, pointing directly to the

versions most in need of further investigation. The results are presented in Attachment 6.

Finally, for deeper analysis, we constructed horizontal barplots tracking sentiment distributions across versions, allowing us to observe recurring issues, shifts in perception, or persistent complaint patterns over time. These visualizations in following **link** provide a broader context that complements the statistical results. From manual examination, several interesting patterns emerge. For Claude, no strongly persistent issue was found, though earlier complaints were often about phone number requirements, later shifting toward usage limits in recent versions. In Gemini, comparisons with Google Assistant appeared as a recurring theme, showing persistence across updates. For DeepSeek, the initial versions were dominated by sign-up related issues, but these gradually shifted toward server busy errors, which became persistent in later versions. In Grok, complaints were highly varied with no single issue persisting across versions. Meanwhile, for GPT, earlier complaints revolved around generated Ghibli-style images, but in later versions, the main issue shifted toward slow response times. Together, these findings highlight that while the Chi-Square and sentiment analyses reveal whether significant differences exist across versions, the complaint-level inspection sheds light on what drives those differences, showing how user concerns evolve, persist, or disappear with each app update.

CHAPTER V: Conclusion and Recommendation

A. Conclusion

This study demonstrates that sentiment classification on imbalanced LLM app reviews can be performed both accurately and fairly. Across experiments, the best results were achieved using twt-roBERTa-stm and Multilingual E5, each reaching an F1-Macro score of 0.66–0.67, clearly outperforming other encoders. Interestingly, despite its lightweight nature, *twt-roBERTa-stm* (with prior sentiment knowledge) performed competitively with larger extractors such as *Qwen3/0.6B*. The choice of classifier (LightGBM, XGBoost, CatBoost, or ANN) had minimal impact on overall performance, reinforcing that the encoder plays the dominant role.

Negative review analysis revealed recurring themes such as usability issues, reliability concerns, and dissatisfaction with core features. These themes were identified through frequent keyword distributions and further contextualized by inspecting the original review sentences. In addition, cross-version sentiment analysis showed clear shifts in user perceptions across application releases. Using Chi-Square tests and standardized residuals, we detected versions with unusually high proportions of negative feedback, which pointed to specific updates that introduced or amplified user complaints.

Taken together, these findings answer the research questions by showing that: (1) sentiment models can reliably capture nuanced, imbalanced feedback with moderate fairness, (2) systematic analysis of negative reviews highlights core user

pain points, and (3) statistical monitoring across versions effectively detects sentiment and issue shifts that developers can act upon.

B. Recommendation

Based on the findings of this study, several recommendations can be drawn for both future research and practical application.

a. Modeling Approach
   - Although large-scale state-of-the-art encoders such as Qwen3 performed competitively, lightweight domain-specific models like *twitter-RoBERTa* and *multilingual E5* proved equally effective, achieving Macro F1-scores of 0.66–0.67. This suggests that researchers and practitioners may prioritize smaller, fine-tuned models for sentiment analysis of user reviews, as they provide strong performance with lower computational cost.
   - Since classifier choice (LightGBM, XGBoost, CatBoost, or ANN) had minimal impact compared to encoder selection, future work should allocate more effort toward exploring and optimizing embedding models rather than complex classifier architectures.

b. Insight Extraction from Negative Reviews
   - Negative review analysis consistently highlighted themes of **usability, reliability, and feature dissatisfaction**. Developers should establish continuous monitoring pipelines that automatically extract and cluster negative feedback, allowing faster detection of recurrent pain points.
   - Contextual inspection remains essential, as keywords alone may not fully capture user intent. Hence, pairing keyword frequency analysis with qualitative review sampling is recommended to prevent misinterpretation of user concerns.

c. Cross-Version Monitoring
   - Sentiment shifts across app versions were statistically confirmed using Chi-Square tests and standardized residuals, revealing that certain updates disproportionately triggered negative reactions. This indicates that incorporating release-aware sentiment monitoring into development cycles is critical.
   - In practice, developers should link spikes in negative sentiment to specific feature rollouts or bug introductions, followed by targeted hotfixes or communication strategies to rebuild user trust.

d. Future Research
   - Expanding beyond sentiment classification, future work can integrate aspect-based sentiment analysis (ABSA) to more precisely capture which features or functions are driving user dissatisfaction.
   - Additionally, incorporating multilingual sentiment adaptation without reliance on external translators could improve consistency, especially for user bases with diverse languages.

REFERENCES

[1] E. Bailyn, Top generative AI chatbots by market share, *First Page Sage*, 2025. [Onine]. URL https://firstpagesage.com/reports/top-generative-ai-chatbots/.

[2] Asfahan, Mengukur popularitas ChatGPT, Gemini, & DeepSeek, *Infografis Bloomberg Technoz,* Jun. 12, 2025. [Online]. URL https://www.bloombergtechnoz.com/detail-infografis/694/mengukur-popularitas-chatgpt-gemini-deepseek.

[3] V. Chhetri, K. Upadhyay, A. B. Siddique, and U. Farooq, What users value and critique: Large-scale analysis of user feedback on AI-powered mobile apps, *arXiv preprint arXiv:2506.10785*, 2025.

[4] D. Martens and T. Johann, On the emotion of users in app reviews, *arXiv preprint arXiv:1703.02256*, 2017.

[5] T. Hou, B. Yannou, Y. Leroy, and E. Poirson, Mining customer product reviews for product development: A summarization process, *Expert Systems with Applications*, vol. 132, pp. 141-150, 2019, doi: 10.1016/j.eswa.2019.04.069.

[6] L. Ashbaugh and Y. Zhang, A comparative study of sentiment analysis on customer reviews using machine learning and deep Learning, *Computers*, vol. 13, no. 12, article 340, Dec. 2024. doi: 10.3390/computers13120340.

[7] M. Aithal and C. Tan, On positivity bias in negative reviews, in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, 2021, pp. 294–304. doi: 10.18653/v1/2021.acl-short.39.

[8] T. Yang, C. Gao, J. Zang, D. Lo, and M. R. Lyu, TOUR: Dynamic topic and sentiment analysis of user reviews for assisting app release, in *Companion Proceedings of the Web Conference 2021 (WWW '21 Companion)*, Ljubljana, Slovenia, Apr. 2021, pp. 724-728. doi: 10.1145/3442442.3458612.

[9] M. A. Hadi and F. H. Fard, AOBTM: Adaptive online biterm topic modeling for version sensitive short-texts analysis, *arXiv preprint arXiv:2009.09930*, 2020.

[10] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, Learning representations by back propagating errors, *Nature*, vol. 323, no. 6088, pp. 533–536, Jan. 1986, doi: 10.1038/323533a0.

[11] Sepp Hochreiter and Jurgen Schmidhuber. Long short-term memory. *Neural computation*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.

[12] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, Attention is all you need, in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 5998-6008.

[13] L. Wang, N. Yang, X. Huang, B. Jiao, L. Yang, D. Jiang, R. Majumder, and F. Wei, Text embeddings by weakly-supervised contrastive pre-training, *arXiv preprint arXiv:2212.03533*, 2022.

[14] Y. Zhang, M. Li, D. Long, X. Zhang, H. Lin, B. Yang, P. Xie, A. Yang, D. Liu, J. Lin, F. Huang, and J. Zhou, Qwen3 embedding: Advancing text embedding and reranking through foundation models, *arXiv preprint arXiv:2506.05176,* 2025.

[15] N. Muennighoff, N. Tazi, L. Magne, and N. Reimers, MTEB: Massive text embedding benchmark, *arXiv preprint arXiv:2210.07316,* 2022.

[16] J. H. Friedman, Greedy function approximation: A gradient boosting machine, *Annals of Statistics,* vol. 29, no. 5, pp. 1189-1232, 2001.

[17] H. Du, Machine learning: decision trees, *Programming Domain*, 2024. [Online]. URL https://programming.earthonline.us/16-machine-learning-decision-trees-b9c2d639 7d8b.

[18] T. Chen and C. Guestrin, XGBoost: A scalable tree boosting system, in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)*, San Francisco, CA, USA, 2016, pp. 785-794, doi: 10.1145/2939672.2939785.

[19] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, LightGBM: A highly efficient gradient boosting decision tree, in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 3147–3155.

[20] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, CatBoost: unbiased boosting with categorical features, *arXiv preprint arXiv:1706.09516,* 2017.

[21] J. Alammar and M. Grootendorst, Hands-on large language models, *Sebastopol, CA, USA: O'Reilly Media, Inc.*, 2025.

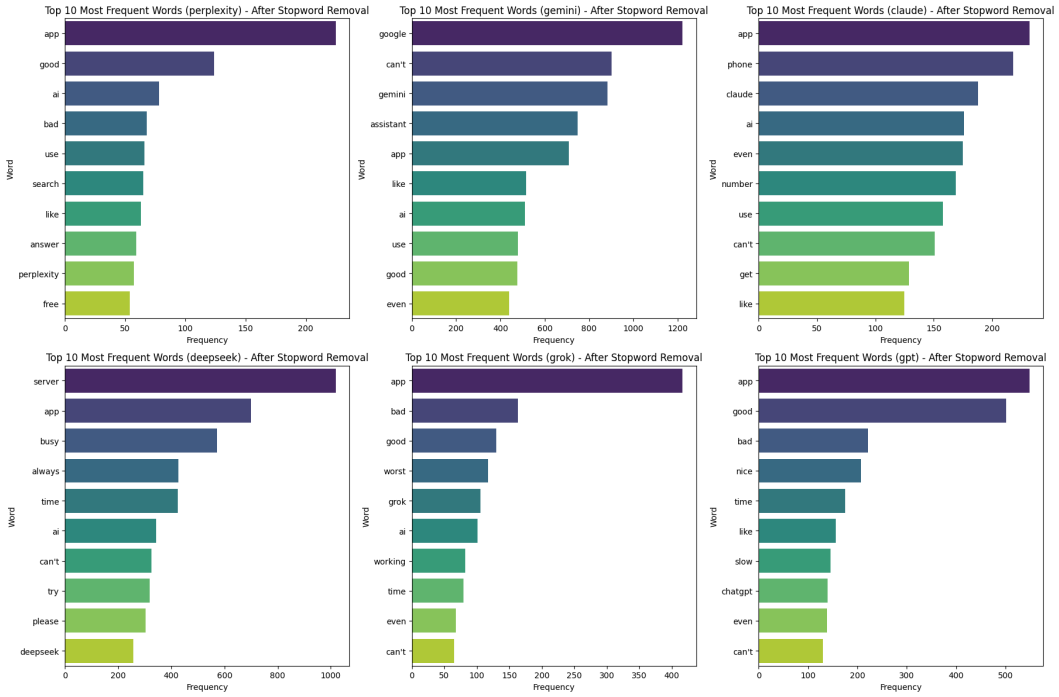
[22] M. Grootendorst, BERTopic: Neural topic modeling with a class-based TF-IDF procedure, *arXiv preprint arXiv:2203.05794*, 2022.

ATTACHMENT

Attachment 1: Snapshot of dataset

Comment	At	AppVersion	Sentiment
Ridiculous how we are limited when paying!	2025-06-10 21:24:17	1.250609.27	2
love this app but would like a download icon to appear in my downloaded apps.	2025-04-05 21:24:04	1.250331.7	2
perfect AI i can't compare with even Chatgpt this is best 🙌🙌🙌🤜🤜🤜🤜🥰🥰🥰🥰🥰🥰	2025-05-30 08:17:46	1.250526.17	2

Attachment 2: Horizontal barplot of most frequent words in negative reviews after stopwords removal



Himpunan Mahasiswa Statistika ITS (HIMASTA-ITS)
Gedung H Lantai III, Jl. Arief Rahman Hakim
Kampus ITS Sukolilo Surabaya
Email: pekanrayastatistika2025@gmail.com
Contact Person:
Hita (+62 857-1207-7572)
Ali (+62 812-3590-7687)

Attachment 3: Sample rows that contained the most frequent words for each LLM

```
Most frequent words in negative reviews for perplexity (excluding common words): ['search', 'answer', 'free', 'time', 'pro']
Sample negative reviews containing these words for perplexity:
```

	Comment_cleaned
121589	ai. current state lightyear behind ai/ml models edit: response devoloper- glad see curious feedback routing endless loop feedback. compared bard chat gpt. think much ask, said hi gave definition usage hi. asked said responds based search engine services. currently tool combined search engine, ai
130504	much censoring ai search happened freedom speech give option setting turn censorship search leave people decide turn censorship uninstalling app censorship search
128756	asked scheduled afl football games twice perplexity gave wrong answers times. also used past tense game played yet. also asked accurate ai "perplexity" app said "based provided search results, unclear "perplexity ai app" refers to." hmm.
119657	totally waste time. fully customized us only. ask anything, gives answer based us data. better use bard ai.
122989	update perplexity app give information part answer shoul be, photos maybe sometimes videos. written explanation.
121174	app provide in-text citations academic work. give high lighted numbers click it, take site, good researcher.
125724	fantastic application ever, giving answers way expect
126765	good. english only, prospect making multilingual already 10 years existence box ... 90% world ignored. weird. :thinking_face:

```
Most frequent words in negative reviews for gemini (excluding common words): ['google', 'assistant', 'want', 'phone', 'voice']
Sample negative reviews containing these words for gemini:
```

	Comment_cleaned
85001	loved bard better competitor new gemini terrible replacement assistant. able anything would normally use assistant click submit even using voice command. can't control flashlight, home devices, reminders, etc. used able hands free assistant. still revert back old assistant can't use ai chat app app gets reset (i pixel.)
74095	terrible experience. try substitute assistant work all. do. gets information train schedules incorrect, balky controlling lights. really see point application uninstalling. bad enough definitely reconsider choice buy android phones future turns future google assistant.
60542	supported s22+? really google??
80313	get together google. gemini cannot assistant actions, access google tasks reason, can't run routines, requires clicking send button. offer replacement assistant point
59571	well, service good, pick getting gemini generative ai responses able turn on/off lights voice. can't assistant voice control access gemini app? use unless also still control devices voice.
69392	lacks ability interact home controls, appear usable without replacing assistant can. without home control integration, kind pointless since functionality available via natural means. ability might helpful, installing strictly removes functionality.
70971	support nothing 2 phone. even possible? text prompt, work android phone.

```
Most frequent words in negative reviews for claude (excluding common words): ['phone', 'number', 'get', 'free', 'error']
Sample negative reviews containing these words for claude:
```

	Comment_cleaned
42540	waiting time long pro user.. used 30 minutes says wait 4 hours, get it! pro user then? want pay need solution right now, wanna wait.. please fix issue asap!
37338	can't register. says phone number invalid.
37675	can't log canada, requires phone number, canadian option :/
43121	entering phone numbr
38261	needs verified email phone? ty, use gpt
39599	accept puerto rico telephone number registration
40201	need phone number?
39556	f need phone number? nope. reason nuked chatgpt app. ain't giving phone number.
42719	support iran's phone number.

```
Most frequent words in negative reviews for deepseek (excluding common words): ['server', 'busy', 'always', 'time', 'try']
Sample negative reviews containing these words for deepseek:
```

	Comment_cleaned
53536	bad much censorship :china: chinese spyware :no_entry:
51369	first fix app bug application working properly smoothly sometimes unable send text caz stick button working
45805	read privacy pilicy.. store information collect secure servers located people's republic china . thsnks!!
45291	study tablet studies, need ai tool. suggested guess rotate. need rotate screen tablet always standing position. rotate laptop vibe comfortablity. want rotate tablet using deep seek better could add rotation app. shocked doens't feature yet
58810	trying interact send messages anything. buggy work. sorry, 1 star.
44835	8 times got code sign always failed, finally uninstall.
47174	trying sign up, anytime enter code keeps rejecting it. saying code invalid expired. sorted, 2 star reward.
45509	try asking tiananmen square

Himpunan Mahasiswa Statistika ITS (HIMASTA-ITS)
Gedung H Lantai III, Jl. Arief Rahman Hakim
Kampus ITS Sukolilo Surabaya
Email: pekanrayastatistika2025@gmail.com
Contact Person:
Hita (+62 857-1207-7572)
Ali (+62 812-3590-7687)

```
Most frequent words in negative reviews for grok (excluding common words): ['working', 'time', 'work', 'chat', 'image']
Sample negative reviews containing these words for grok:
```

	Comment_cleaned
113477	nothing differential tier 2 3 ais. chat gpt gemeni still best.
117370	average working grok beta
106606	supposed make voice chat option would happy rate it.
117925	difference app chat gpt using chatgpt need use app extra use phone space chatgpt even gives free question answer ask questions free 3 times worth use app. choice opinion....
105967	saw screenshot try, unable create image, grok told text based ai. :neutral_face:
105755	understanding files images properly
99133	images per prompt
97622	review: using grok app information searching discovered valuable insights. however, posing complex questions, tab suddenly closed. repeated attempts reopen result app crashing. incredibly frustrating invested significant time gathered information lost. appears recovery option. lack data recovery app malfunction major issue.

```
Most frequent words in negative reviews for gpt (excluding common words): ['time', 'slow', 'image', 'work', 'working']
Sample negative reviews containing these words for gpt:
```

	Comment_cleaned
6044	bad app working
17923	slow producting ghibi art image
21872	slow :confused_face:
16169	keep saying update playstore although latest,it's working shall
7659	slow :sloth:
20438	slow :sloth::disappointed_face:
25552	app working properly
19738	work realme 10 pro ??
5749	slow super :pouting_cat::pouting_cat::pouting_cat:

Attachment 4: Rows with missing values (rows already in chronological order)

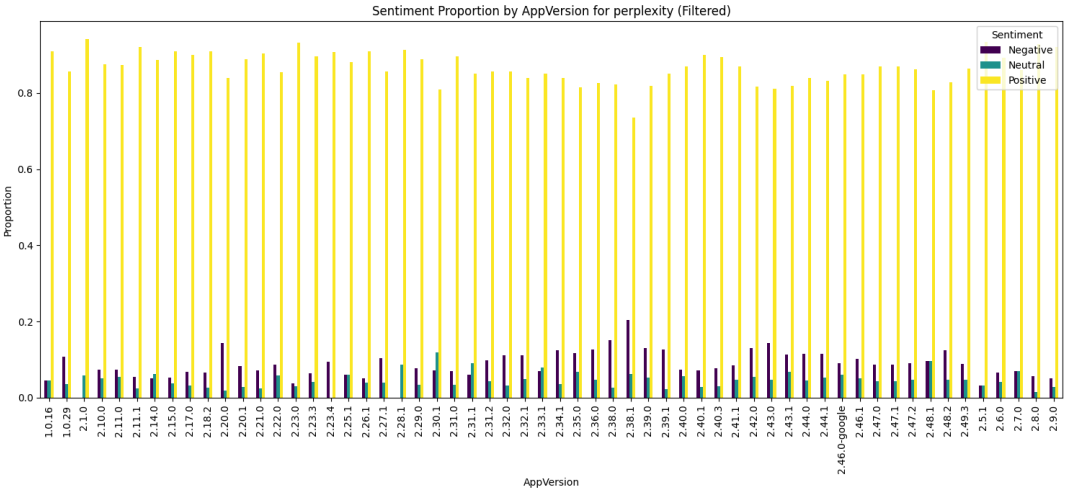
120109	2023-05-30 20:21:40	1.0.16	2	perplexity	it's an honor to be one of the first downloade...
124297	2023-05-30 20:40:35	NaN	2	perplexity	best ui design i have ever seen fr
118973	2023-05-30 21:25:06	1.0.16	2	perplexity	best app so far, congrats team keep it going
86479	2024-02-08 15:23:51	1.0.603736800	2	gemini	good start, but the integration with assistant...
74095	2024-02-08 15:35:10	NaN	0	gemini	terrible experience. if you try to substitute ...
60542	2024-02-08 15:38:26	1.0.603736800	0	gemini	not supported on s22+? really google??
39044	2024-07-16 07:33:04	1.240722.0	2	claude	i like this app, it makes my life a little eas...
41760	2024-07-16 11:10:31	NaN	2	claude	phenomenal ai. great ui, very nice of claude h...
42941	2024-07-16 15:08:57	1.240722.0	2	claude	the app is exactly what i expected. very clean...
47794	2025-01-15 10:16:31	1.0.2	2	deepseek	perfect
46154	2025-01-15 10:44:24	NaN	2	deepseek	good
45607	2025-01-15 11:01:07	1.0.2	2	deepseek	i love it. good jobs developers. keep up the g...
117969	2025-02-27 08:50:14	0.1.17	2	grok	yay look forward to more features!!!
111892	2025-02-27 08:57:50	NaN	2	grok	amazing app thanks elon musk
117195	2025-02-27 09:11:04	0.1.17	2	grok	uncensored, unwoked, very very good answers.

Himpunan Mahasiswa Statistika ITS (HIMASTA-ITS)
Gedung H Lantai III, Jl. Arief Rahman Hakim
Kampus ITS Sukolilo Surabaya
Email: pekanrayastatistika2025@gmail.com
Contact Person:
Hita (+62 857-1207-7572)
Ali (+62 812-3590-7687)

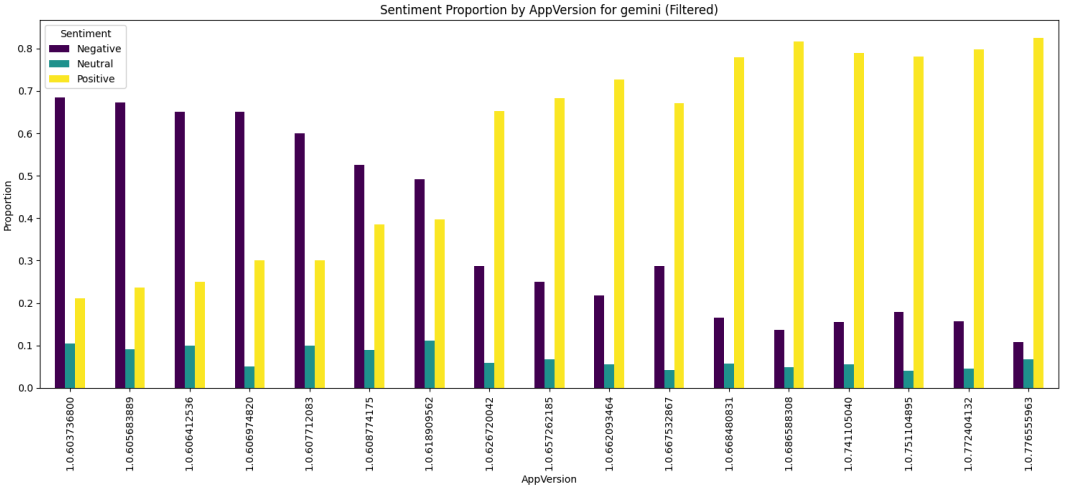
6940	2025-07-08 02:55:41	1.2025.182	2	gpt	better than human fake support
699	2025-07-08 02:56:02	NaN	2	gpt	this is good and he helps other people to unde...
9613	2025-07-08 02:57:44	1.2025.182	2	gpt	informative

Attachment 5: Sentiment proportion across app-versions

(a.) Perplexity

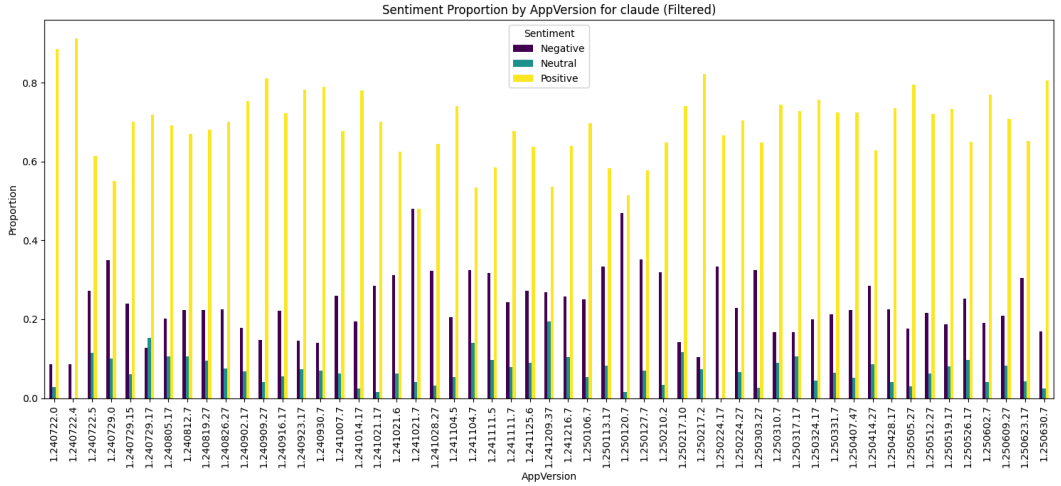


(b.) Gemini

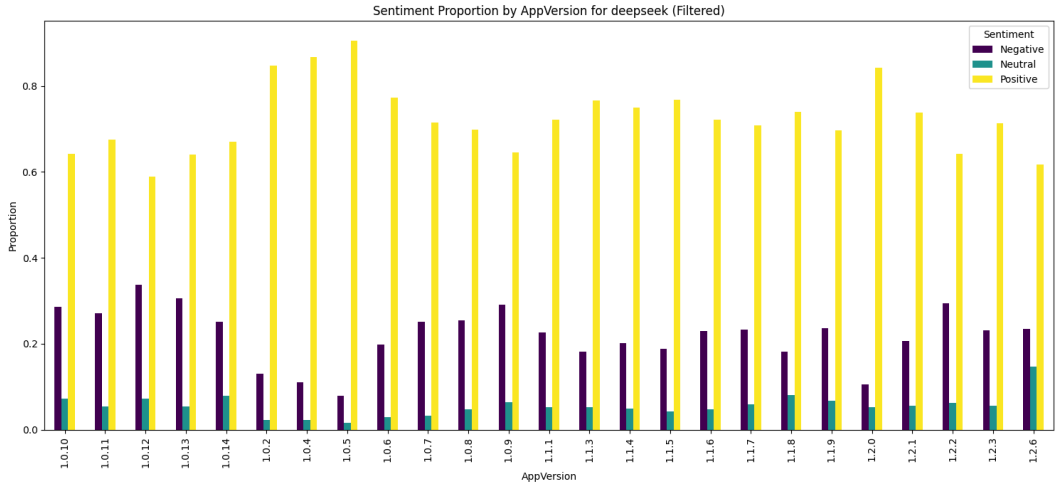


Himpunan Mahasiswa Statistika ITS (HIMASTA-ITS)
Gedung H Lantai III, Jl. Arief Rahman Hakim
Kampus ITS Sukolilo Surabaya
Email: pekanrayastatistika2025@gmail.com
Contact Person:
Hita (+62 857-1207-7572)
Ali (+62 812-3590-7687)

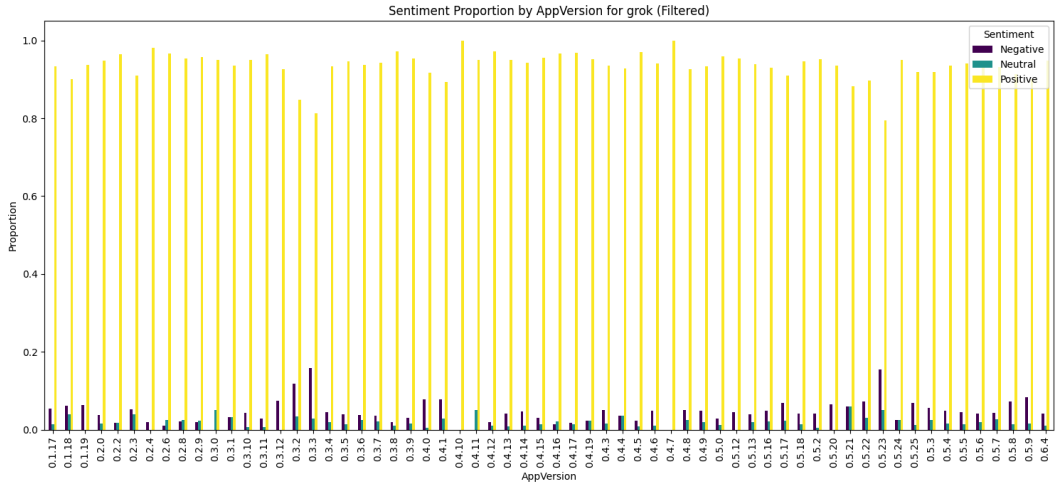
(c.) Claude



(d.) Deepseek

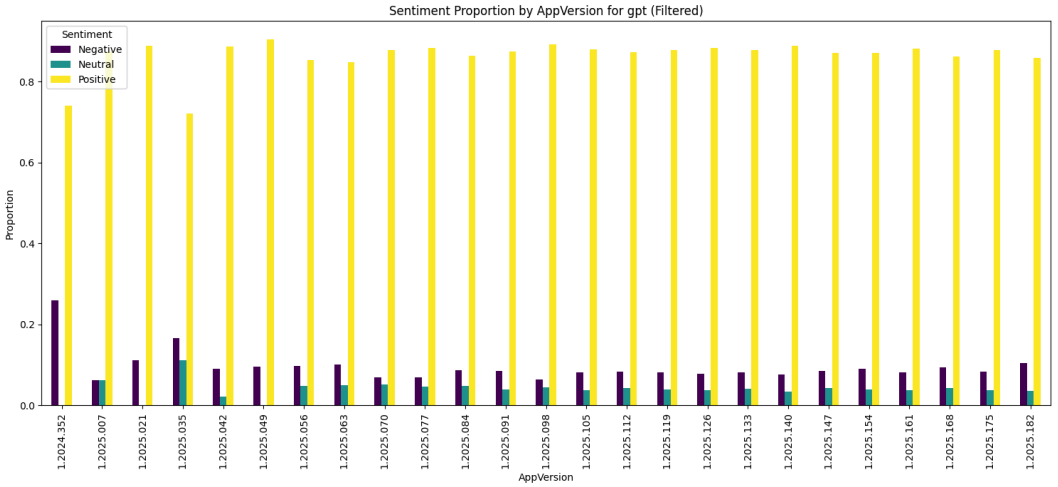


(e.) Grok



Himpunan Mahasiswa Statistika ITS (HIMASTA-ITS)
Gedung H Lantai III, Jl. Arief Rahman Hakim
Kampus ITS Sukolilo Surabaya
Email: pekanrayastatistika2025@gmail.com
Contact Person:
Hita (+62 857-1207-7572)
Ali (+62 812-3590-7687)

(f.) GPT



Attachment 6: Standardized Residuals by App Version

LLM App	App Version	Sentiment	Residual	Interpretation
Perplexity	2.15.0	Negative	-2.26	Fewer negatives than expected
	2.30.1	Neutral	2.28	More neutrals than expected
	2.42.0	Negative	3.38	More negatives than expected
	2.9.0	Negative	-3.90	Fewer negatives than expected
	2.9.0	Neutral	-2.39	Fewer neutrals than expected
Gemini	1.0.608774175	Negative	14.25	Far more negatives than expected
	1.0.608774175	Positive	-7.90	Far fewer positives than expected
	1.0.686588308	Negative	-13.07	Far fewer negatives than expected
	1.0.686588308	Positive	6.74	More positives than expected

Claude	1.241021.7	Negative	5.93	More negatives than expected
	1.241021.7	Positive	-2.97	Fewer positives than expected
	1.250217.2	Negative	-3.29	Fewer negatives than expected
Deepseek	1.0.12	Negative	3.64	More negatives than expected
	1.0.12	Positive	-2.67	Fewer positives than expected
	1.0.6	Neutral	-2.39	Fewer neutrals than expected
Grok	0.3.2	Negative	14.26	Far more negatives than expected
	0.3.2	Positive	-4.36	Fewer positives than expected
	0.3.3	Negative	18.88	Far more negatives than expected
	0.3.3	Positive	-5.07	Fewer positives than expected
GPT	1.2024.352	Negative	3.13	More negatives than expected
	1.2025.098	Negative	-2.62	Fewer negatives than expected

Himpunan Mahasiswa Statistika ITS (HIMASTA-ITS)
Gedung H Lantai III, Jl. Arief Rahman Hakim
Kampus ITS Sukolilo Surabaya
Email: pekanrayastatistika2025@gmail.com
Contact Person:
Hita (+62 857-1207-7572)
Ali (+62 812-3590-7687)